

Confidence Intervals (Part I)

STAT 24510-30040

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Confidence interval is the most informative among classical parameter inference methods of parameter estimation, hypothesis testing, and confidence interval. We will investigate several methods of building confidence intervals using examples, represented by important, common probability distributions.

1 Review of point estimation and confidence intervals

Point estimation review

- θ , a parameter of interest.
- $\hat{\theta} = \hat{\theta}(x)$, an estimate of θ from observed data x by a procedure, such as Least Squares or Maximum Likelihood.
- The estimate should be paired with a measure of its accuracy, usually its standard error.

Common notations and abbreviations:

- *s.e.* or *SE*: standard error, for an estimator.
- *i.i.d.*: Independent, identically distributed, for a sequence of random variables.
- $N(\mu, \sigma^2)$: the probability distribution of a normal random variable with mean μ and variance σ^2 .
- Conventionally, given observed data x ,
 - an estimate $\hat{\theta}(x)$ is a realization of an estimator $\hat{\theta}(X)$ when $X = x$, where
 - X represents the unknown random sample drawn from a population, and
 - $\hat{\theta}(X)$ formulates an estimation procedure, such as
 - * the maximum likelihood estimator (MLE)
 - * the estimator by the method of moments (MoM)
 - * the Least Squares Estimator (MLE)

Example:

Data points $x_1, \dots, x_n$ are often viewed as independent observations from a common probability distribution F_θ , where $\theta \in \mathbb{R}$ is an unknown parameter of interest. The most common example of θ is the population mean.

$s.e.(\hat{\theta})$ = standard error of $\hat{\theta}$, which is the estimated standard deviation of $\hat{\theta}$.

If $\hat{\theta} = 15.4$ is unbiased, $s.e.(\hat{\theta}) = 1.7$, and if $\hat{\theta}$ is approximately of normal distribution, then

the probability that 15.4 is within one *s.e.* of θ is about 0.68 or 68%; within 2 *s.e.*'s about 0.95.

Heuristic review of the confidence interval concept

An alternative way to point estimation is using confidence intervals:

Find two numbers L, U from data, such that the interval $[L, U]$ covers θ with some "level of confidence".

Properties:

- " $[\hat{L}, \hat{U}]$ is a 95% confidence interval for θ " is an interval estimate.
 - 95% is the probability that the **random** interval captures the **fixed** parameter.
 - The probability that such random interval $[\hat{L}, \hat{U}]$ contains the true value of θ is 0.95.
- $[\hat{L}, \hat{U}]$ can be calculated from data.
 - For example, a statement based on observed data can be "The interval (12.0, 18.8) is an approximate 95% confidence interval for the parameter θ ."
- Narrower confidence intervals are more useful, thus preferred.
- Advantages of confidence intervals: Combining an assessment of accuracy within the estimates.
 - Disadvantages of confidence intervals: Prone to misinterpretation and model misspecification.
- Bayesian approach:
 - Make an *a priori* statement about the probability distribution of θ , then obtain a posteriori statement about the probability distribution of θ based on data.

2 The pivotal method for confidence intervals

Pivotal methods construct confidence intervals using a type of function $T = T(X, \theta)$ of data X and parameter θ , such that T has a probability distribution free of the parameter θ . The parameter-free probability distribution of $T = T(X, \theta)$ then can be converted to a statement of the accuracy of an estimator $\hat{\theta}(X)$ from the true value of θ .

Definition

A **pivotal** quantity T is a function of random variables $X_1, \dots, X_n$ and unknown parameter θ such that the probability distribution of T does not depend on θ .

Such T can be used to construct so-called **exact** confidence intervals for θ , using the pivotal method.

In this context, “exact” means that the confidence interval has exactly the stated confidence level (e.g., 95%) for the given sample size, as opposed to asymptotic or large-sample confidence intervals which only have approximately the stated confidence level (e.g., $\approx 95\%$) and only when the sample size is sufficiently large.

Construction of exact confidence intervals by the pivotal method

Suppose the test level is α (e.g., $\alpha = 0.05$).

That is, suppose the desired level of confidence of the confidence interval is $(1 - \alpha)100\%$.

To construct a confidence interval by pivotal method:

(a) Find a pivotal quantity $T = T(X, \theta)$ such that the distribution of T does not depend on the parameter θ .

(b) Find numbers (called critical values) c_L and c_U from the probability distribution of T such that

$$\mathbb{P}(c_L < T < c_U) = 1 - \alpha$$

where $1 - \alpha$ is the desired confidence level.

Note: The end points can be either open or closed, for example, it can be $\mathbb{P}(c_L \leq T \leq c_U) = 1 - \alpha$.

(c) Solve the inequalities to obtain the confidence set for θ :

$$\{\theta : c_L < T(X, \theta) < c_U\}$$

Ideally, the confidence set is an interval, and the two random variables L, U (not depending on θ) can be calculated from data so that

$$\mathbb{P}(L < \theta < U) = 1 - \alpha$$

This probability is called **coverage probability**.

First, let's walk through the construction steps of the pivotal method by a simple example.

Example 1 (Normal location parameter)

Data: $X_1, \dots, X_n$.

Model assumption: X_i 's are *i.i.d.* of normal distribution with mean θ and variance 25, denoted as $X_i \sim N(\theta, 25)$.

Goal: Estimate the location parameter θ , the population mean of the X_i 's, then construct a confidence interval.

The maximum likelihood estimator (MLE) of θ is the sample mean, $\hat{\theta}_{ML} = \bar{X} \sim N(\theta, \frac{25}{n})$.

In the following we show that, we can find two random variables L and U not depending on θ such that the probability

$$\mathbb{P}(L \leq \theta \leq U) = 1 - \alpha.$$

(a) Choose

$$T = \frac{\bar{X} - \theta}{\sqrt{25/n}} \Rightarrow T \sim N(0, 1)$$

$T = T(X_1, \dots, X_n, \theta)$ is a function of data $\{X_1, \dots, X_n\}$ and the parameter θ .

Note that the **probability distribution of T does not depend on the parameter θ** .

Such T (with a parameter-free distribution) is called a **pivotal**, or a pivotal quantity.

(b) Next, find critical values c_L and c_U from the probability distribution of T such that

$$\mathbb{P}(c_L \leq T \leq c_U) = 1 - \alpha$$

There are infinitely many pairs of (c_L, c_U) satisfying the above inequality to yield the same probability $1 - \alpha$.

If the pivotal probability distribution is symmetric (such as the normal distribution here), the optimal confidence interval is the narrowest one, which is also equal tailed.

Let z_α denote the α th lower-quantile of $N(0, 1)$ with $\mathbb{P}(Z \leq z_\alpha) = \alpha$, where $Z \sim N(0, 1)$ is a standard normal random variable. For example $\alpha = 0.025$ then $z_\alpha = -1.96$.

Since $T \sim N(0, 1)$, by the symmetry of the probability density function of $N(0, 1)$,

$$\mathbb{P}(z_{\alpha/2} \leq T \leq z_{1-\alpha/2}) = 1 - \alpha$$

where the symmetric quantiles provide the narrowest bounds for T .

This is a probability on the deterministic bounds of a random variable T which depends on data via $\bar{X}$ and unknown parameter θ .

(c) In the following we work on converting the probability to be a statement about the plausibility of θ .

Ideally, two random variables L, U (not depending on θ) can be found such that

$$\mathbb{P}(L \leq \theta \leq U) = 1 - \alpha$$

For this example, we can restate the inequalities inside the probability in terms of $\bar{X}$,

$$\mathbb{P}(z_{\alpha/2} \leq T \leq z_{1-\alpha/2}) = \mathbb{P}\left(z_{\alpha/2} \leq \frac{\bar{X} - \theta}{\sqrt{25/n}} \leq z_{1-\alpha/2}\right) = \mathbb{P}\left(z_{\alpha/2} \frac{5}{\sqrt{n}} \leq \bar{X} - \theta \leq z_{1-\alpha/2} \frac{5}{\sqrt{n}}\right) = 1 - \alpha$$

Multiplying each term in the last probability expression by -1 reverses the inequalities inside the probability,

$$-z_{1-\alpha/2} \frac{5}{\sqrt{n}} \leq \theta - \bar{X} \leq -z_{\alpha/2} \frac{5}{\sqrt{n}}$$

By the symmetric property of normal distribution, $z_{1-\alpha/2} = -z_{\alpha/2} \geq 0$. The inequalities become

$$-z_{1-\alpha/2} \frac{5}{\sqrt{n}} \leq \theta - \bar{X} \leq z_{1-\alpha/2} \frac{5}{\sqrt{n}} \iff \bar{X} - z_{1-\alpha/2} \frac{5}{\sqrt{n}} \leq \theta \leq \bar{X} + z_{1-\alpha/2} \frac{5}{\sqrt{n}}$$

Another way to get the above expression from $z_{\alpha/2} \frac{5}{\sqrt{n}} \leq \bar{X} - \theta \leq z_{1-\alpha/2} \frac{5}{\sqrt{n}}$ is to solve the two inequalities $z_{\alpha/2} \frac{5}{\sqrt{n}} \leq \bar{X} - \theta$ and $\bar{X} - \theta \leq z_{1-\alpha/2} \frac{5}{\sqrt{n}}$ then combine the results.

These algebraic operations do not change the random event. The resulting probability equation has the desired form,

$$\mathbb{P}(L \leq \theta \leq U) = \mathbb{P}\left(\bar{X} - z_{1-\alpha/2} \frac{5}{\sqrt{n}} \leq \theta \leq \bar{X} + z_{1-\alpha/2} \frac{5}{\sqrt{n}}\right) = 1 - \alpha$$

Now the original probability statement on fixed bounds of random statistic T is re-expressed as a probability statement on the random bounds of parameter θ , which provides a confidence interval for θ .

This re-expression demonstrates why the meaning of confidence interval is stated is such a round-about way.

We say that

$$[L, U] = \left[\bar{X} - z_{1-\alpha/2} \frac{5}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{5}{\sqrt{n}} \right]$$

is a $(1 - \alpha)100\%$ confidence interval for the parameter θ , where L and U are random variables depending on the data through $\bar{X}$. We say the confidence interval has $1 - \alpha$ **coverage probability**.

The meaning of coverage probability or confidence level

If we could collect such samples (each sample consisting of n data points) K many times (e.g. $K = 500$), we compute $\bar{X}$ for each sample, and obtain a $(1 - \alpha)100\%$ confidence interval for θ from each sample, then there would be around $(1 - \alpha) \times K$ of the confidence intervals actually containing the true θ , while $\alpha \times K$ of the K confidence intervals would not contain the true θ .

If we use test level $\alpha = 0.05$, a common choice, then the upper quantile $z_{1-\alpha/2} = 1.96$.

We then obtain a 95% exact confidence interval $[L, U]$ for parameter θ ,

$$\mathbb{P}(L \leq \theta \leq U) = \mathbb{P}\left(\bar{X} - 1.96 \frac{5}{\sqrt{n}} \leq \theta \leq \bar{X} + 1.96 \frac{5}{\sqrt{n}}\right) = 0.95$$

As (we imagine) $\bar{X}$ varies from sample to sample, the **random interval**

$$[L, U] = \left[\bar{X} - 1.96 \frac{5}{\sqrt{n}}, \bar{X} + 1.96 \frac{5}{\sqrt{n}} \right]$$

contains the **fixed** parameter θ with coverage probability 0.95; i.e., $[L, U]$ is a 95% confidence interval for θ .

Remarks

- End points of the confidence intervals:

In this example, the random variables X_i 's are of continuous distribution, therefore $\mathbb{P}(X_i = x_0) = 0$ at any single point $x_0 \in (-\infty, \infty)$. Since L and U are continuous functions of the X_i 's, they have the same property of zero probability at any discrete point. Therefore the open interval (L, U) and the closed interval $[L, U]$ have the same confidence level or the same coverage probability.

However, (L, U) and $[L, U]$ may have different coverage probabilities for samples from discrete distributions.

- Effects of sample size on the width of the confidence interval:

For fixed test level α , the larger the sample size n , the narrower the confidence interval.

As $n \rightarrow \infty$, the length of the interval $\rightarrow 0$. (This is a desirable property, called the consistency of the estimator.)

- Effects of confidence level on the width of the confidence interval:

For a given sample size n , the smaller the test level α , the wider the confidence interval.

As $\alpha \rightarrow 0$, the length of the interval $\rightarrow \infty$.

The pivotal method can be used to derive confidence intervals for parameters other than the mean, such as the variance parameter, as shown in the following example.

Example 2 (Normal scale parameter)

Data: $X_1, \dots, X_n$.

Model assumption: X_i 's are *i.i.d.* with $X_i \sim N(3, \sigma^2)$, σ is unknown.

Goal: Estimate the variance parameter σ^2 and construct a confidence interval for σ^2 .

First, recall the fact that, if Z_i 's are *i.i.d.* of common distribution $N(0, 1)$, then the probability distribution of $Z_1^2 + \dots + Z_k^2$ is named as chi-square distribution with a parameter k , termed the degrees of freedom. The chi-square distribution with degrees of freedom k is denoted as χ_k^2 or $\chi_{(k)}^2$. We will study this important distribution in more detail later.

In this example, the transformed variable

$$Y_i = \frac{X_i - 3}{\sigma} \sim N(0, 1)$$

and Y_i 's are independent. By the definition of χ^2 distribution, $\sum_{i=1}^n Y_i^2 \sim \chi_n^2$.

We obtain a pivotal quantity

$$T = \frac{\sum_{i=1}^n (X_i - 3)^2}{\sigma^2}, \quad T \sim \chi_n^2.$$

T is a function of data and the parameter σ^2 only, and the distribution of T does not depend on the parameter σ^2 . Thus T is a pivotal quantity, and the probability distribution of T can be used to construct confidence intervals.

Unlike the normal distribution, the pivotal distribution χ_n^2 is skewed with a longer right tail, so its probability density function is not symmetric. The best way to construct confidence interval is no longer obvious.

One reasonable way (though not the only way) to obtain a confidence interval from a asymmetric distribution is to choose equal-tailed critical values.

For $T \sim \chi_n^2$, the critical quantile $\chi_{n,\beta}^2$ of the χ_n^2 distribution can be defined by

$$\mathbb{P}(T \leq \chi_{n,\beta}^2) = \beta$$

So an $(1 - \alpha)100\%$ coverage probability statement can be written as

$$\mathbb{P}\left(\chi_{n,\alpha/2}^2 \leq \frac{\sum_{i=1}^n (X_i - 3)^2}{\sigma^2} \leq \chi_{n,1-\alpha/2}^2\right) = 1 - \alpha$$

To express the inequalities inside the probability as a statement of plausibility of σ^2 in the form $L \leq \sigma^2 \leq U$, note that the probability can be written as

$$\mathbb{P}\left(\frac{1}{\chi_{n,1-\alpha/2}^2} \leq \frac{\sigma^2}{\sum_{i=1}^n (X_i - 3)^2} \leq \frac{1}{\chi_{n,\alpha/2}^2}\right) = 1 - \alpha$$

We then obtain an $(1 - \alpha)100\%$ exact confidence interval for σ^2 ,

$$[L, U] = \left[\frac{\sum_{i=1}^n (X_i - 3)^2}{\chi_{n-1, 1-\alpha/2}^2}, \frac{\sum_{i=1}^n (X_i - 3)^2}{\chi_{n, \alpha/2}^2} \right]$$

Just as in the previous example, L and U are random variables depending on the data X_i 's only. L and U would vary from sample to sample. The random interval $[L, U]$ contains the fixed parameter σ^2 with probability $1 - \alpha$.

Analogously to the previous example, because L and U are continuous functions of continuous random variables, the open interval (L, U) and closed interval $[L, U]$ contain the fixed parameter σ^2 with the same coverage probability.

Example 3 (Normal scale parameter with unknown location parameter)

Data: $X_1, \dots, X_n$.

Model: X_i 's are independent $N(\mu, \sigma^2)$, both μ and σ are unknown.

Goal: Estimate the variance parameter σ^2 and provide a confidence interval for σ^2 .

Since μ is unknown, to estimate σ^2 , it is reasonable to replace $\mu = 3$ in the statistic T in the previous example by the sample mean $\bar{X}$. Then the test statistic becomes

$$T = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$$

Claim T is of χ^2 distribution with $n - 1$ degrees of freedom:

$$T \sim \chi_{n-1}^2$$

(Note: The proof of this claim will be provided later.)

Again T is a function of the data X_i 's and the parameter of interest σ^2 only, and the distribution of T is free of the parameter σ^2 . Therefore, T is a pivotal quantity which can be used to construct confidence intervals.

Notice that by using $\bar{X} = \sum_{i=1}^n X_i/n$ instead of the value of μ , the degrees of freedom of χ^2 is $n - 1$ instead of n . The coverage probability can be stated as

$$\mathbb{P} \left(\chi_{n-1, \alpha/2}^2 \leq \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \leq \chi_{n-1, 1-\alpha/2}^2 \right) = 1 - \alpha$$

Rewrite the inequalities inside the probability as random bounds of σ^2 , we obtain an $(1 - \alpha)100\%$ exact confidence interval for σ^2 ,

$$\left(\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{n-1, 1-\alpha/2}^2}, \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{n-1, \alpha/2}^2} \right).$$

Example 4 (Normal location parameter estimation with variance unknown)

This is the most important example in this section due to its wide applications.

It often happens that there are more than one parameters in the probability distributions in our model, of which only one parameter is the interest of inference (e.g., finding its estimator and and confidence intervals). However we need to deal with parameters in the model that are not of direct interest (called nuisance parameters), as shown in this example (as well as the previous example).

Data: $X_1, \dots, X_n$.

Mode assumption: X_i 's are *i.i.d.* with $X_i \sim N(\theta, \sigma^2)$, both θ and σ are unknown.

Goal: Estimate the location parameter θ and provide a confidence interval for θ .

Recall the pivotal $T = T(X, \theta)$ in the first example of estimating normal mean was

$$T = \frac{\bar{X} - \theta}{\sqrt{\sigma^2/n}}$$

when $\sigma^2 = 25$ is known. But now, T is also a function of the unknown σ^2 , thus this T is no longer a pivotal quantity.

The estimator and the bounds of the confidence interval for θ should be functions of data only. Even though we are only interested in the inference of θ , we still need to replace σ^2 by an estimator from data.

One of the reasonable choice of estimators of σ^2 is the sample variance,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

The main advantage of S^2 is its unbiasedness as an estimator, $\mathbb{E}(S^2) = \sigma^2$ (exercise).

Claim (Theorem) The resulting statistic $T = \frac{\bar{X} - \theta}{\sqrt{S^2/n}}$ is of t distribution with $n - 1$ degrees of freedom.

$$T = \frac{\bar{X} - \theta}{\sqrt{S^2/n}} \sim t_{n-1}$$

Both $\bar{X}$ and S^2 are functions of data only. Given the sample size n , the t distribution with $n - 1$ degrees of freedom (denoted as t_{n-1}) is free of parameters. Therefore this T is a pivotal quantity now.

For test level α , denote $t_{n-1, \alpha/2}$ as the $\alpha/2$ lower-quantile of t_{n-1} ,

$$\mathbb{P}(t_{n-1} \leq t_{n-1, \alpha/2}) = \alpha/2$$

The density function of t is bell-shaped and symmetric with respect to 0, similar to the density of standard normal distribution. The probability statement about T now can be written as a statement about the plausibility of θ ,

$$\mathbb{P} \left(\bar{X} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \leq \theta \leq \bar{X} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \right) = 1 - \alpha$$

The probability represents an exact $(1 - \alpha)\%$ confidence interval for θ , with

$$[L, U] = \left[\bar{X} - t^* \frac{S}{\sqrt{n}}, \bar{X} + t^* \frac{S}{\sqrt{n}} \right]$$

Notes

In the above, we made a claim that

$$\frac{\bar{X} - \theta}{\sqrt{S^2/n}} \sim t_{n-1}$$

when X_i 's are *i.i.d.* with $X_i \sim N(\theta, \sigma^2)$. In fact, this is an important theorem (the theorem of sampling distribution), in which an essential part is making use of the independence of $\bar{X}$ and S^2 , and the fact that $(n - 1)S^2/\sigma$ is of χ^2 distribution with $n - 1$ degrees of freedom.

The derivations for this theorem will be provided later.

Remarks (on confidence intervals by the pivotal method)

- The pivotal method provides a possible procedure to construct exact confidence intervals, in the sense that the confidence intervals have exactly the stated confidence level, that is, the exact probability that the random interval covers the true parameter.
 - Note that the exactness crucially relies on the assumptions, such as the probability distributions and the independence of the sample, which in practice could only be approximately true at best.
 - When the pivotal distribution is not symmetric, there are various criteria to derive confidence intervals. For example, the confidence interval with equal-tail-probabilities ($\alpha/2$ each) is a simple, reasonable choice. However, the equal-tail-probability confidence interval is not necessarily optimal by certain criteria. For example, an equal-tail-probability confidence interval is usually not the narrowest confidence interval.
- can you think of other reasonable ways to construct χ^2 C.I., other than the equal-tailed method we used in the previous two examples?
- In previous examples, X_i 's are of continuous random variables, therefore the open interval (L, U) and the closed interval $[L, U]$ have the same confidence level or the same coverage probability. This is not true for discrete distribution, where (L, U) and $[L, U]$ likely have different coverage probabilities.
 - A side note of clarification: The method of rewriting the inequalities of a probability statement obtained from sample random variables is also used in deriving a type of statistics inference method for parameters, called the fiducial method. However the point of view and the objectives of the fiducial inference are very different from the pivotal method of constricting (frequentist) confidence intervals.
 - Pivot quantity can be hard to obtain in most practical cases, as shown in the following example.

Example 5 (Parameter of discrete Bernoulli and Binomial)

Data: $X_1, \dots, X_n$.

Model: X_i 's are independent Bernoulli random variable, that is,

$$X_i = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1 - p \end{cases}$$

Goal: Estimate the parameter p and provide an confidence interval by the pivotal method (if feasible).

The maximum likelihood estimator for p is $\hat{p} = \bar{X}$, with

$$\mathbb{E}(\hat{p}) = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

We would like to covert a probability statement for a pivotal quantity statistic into the form

$$\mathbb{P}(L \leq p \leq U) = 1 - \alpha$$

where L, U depend on observed data but do not depend on p , so that we can construct confidence interval $[L, U]$ for p from data. However, the statistic

$$\frac{\hat{p} - \mathbb{E}(\hat{p})}{\sqrt{\text{Var}(\hat{p})}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

has a probability distribution depending on the parameter p , thus T is not a pivotal quantity.

This is a case that the usual way to construct a pivotal quantity does not apply.

Let's try another way: construct a confidence interval from probability mass function.

Since our goal is to find L, U such that $\mathbb{P}(L \leq p \leq U) = 1 - \alpha$, rewrite

$$\mathbb{P}(L \leq p \leq U) = \mathbb{P}(nL \leq np \leq nU)$$

We know

$$X_1 + \dots + X_n \sim \text{Binomial}(n, p)$$

The random variables L, U should satisfy

$$\mathbb{P}(nL \leq np \leq nU) = \sum_{k=\lfloor nL \rfloor}^{\lfloor nU \rfloor} \binom{n}{k} p^k (1-p)^{n-k}$$

where the floor function $\lfloor x \rfloor$ denotes the largest integer $\leq x$, and the ceiling function $\lceil x \rceil$ is the smallest integer $\geq x$.

For (at least some) given α , we would like to find expressions of L, U satisfying

$$\sum_{k=\lfloor nL \rfloor}^{\lfloor nU \rfloor} \binom{n}{k} p^k (1-p)^{n-k} = 1 - \alpha$$

It would be very hard to solve for L and U based on this equation, for any $\alpha \in (0, 1)$.

To reduce the complexity, consider approximating the unknown p by the common "plug-in" method (justifiably, by the Law of Large Numbers), that is, using observed sample mean $\hat{p} = \bar{X}$ in the place of the parameter p , we would try to derive estimates $\hat{L}, \hat{U}$ from data by solving

$$\sum_{k=\lfloor n\hat{L} \rfloor}^{\lfloor n\hat{U} \rfloor} \binom{n}{k} \hat{p}^k (1-\hat{p})^{n-k} = 1 - \alpha$$

Now $\hat{L}, \hat{U}$ are the only unknowns in the equation, given observed data. In other words, $\hat{L}, \hat{U}$ are functions of data only.

Therefore, solving for $\hat{L}, \hat{U}$ is possible, but still difficult.

By using $\hat{p}$ in the place of p , we are already using approximation method (via the Law of Large Numbers).

In addition, not all α is solvable due to the discreteness of binomial random variable.

We need to go beyond pivotal methods to construct confidence intervals.

Next, we consider approximation methods, especially, asymptotic methods when the sample size is sufficiently large.

3 Asymptotic method for confidence intervals

In most cases, finding an interval $[L, U]$ with exact $\mathbb{P}(L < \theta < U) = 1 - \alpha$ is difficult or impossible to be achieved, as demonstrated in Example 5 above. If the sample size is sufficiently large, we can use results from the asymptotic theory.

The theoretical results for large samples, based on the Law of Large Numbers and the Central Limit Theorem, are called asymptotic theorems.

For confidence intervals, the asymptotic methods provide approximations

$$\mathbb{P}(L < \theta < U) \approx 1 - \alpha$$

when sample size n is sufficiently large, albeit how large is large enough depends on the problem and the data.

The derivation of the approximation uses Central Limit Theorem and its variations, especially, Slutsky's Theorem.

In the following, we state the most relevant theoretical results used to obtain asymptotic confidence intervals for the sample mean.

Asymptotic normality for sample mean

Assume that $X_1, X_2, \dots$ are independent identically distributed random variables with mean μ and finite variance $\sigma^2 \in (0, \infty)$. Denote the sample mean as $\bar{X}_n = \sum_{i=1}^n X_i/n$.

Note that X_i 's may not be of normal distribution.

The Central Limit Theorem (CLT) says that, for n sufficiently large, the standardized random variable $\frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}}$ is approximately of standard normal distribution.

In the language of convergence of random variables, the asymptotic normality of the CLT can be stated as that, as $n \rightarrow \infty$, the sequence of random variables $\frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}}$ converges in distribution to a standard normal random variable:

$$\lim_{n \rightarrow \infty} F_n(x) = \Phi(x), \quad \forall x \in (-\infty, \infty).$$

where $\forall x \in (-\infty, \infty)$ means for any and all x on the real line, and

$$F_n(x) = \mathbb{P}\left(\frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \leq x\right), \quad \Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz.$$

The same statement can be symbolically written as

$$\frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \xrightarrow{\mathcal{D}} Z \text{ as } n \rightarrow \infty, \quad Z \sim N(0, 1)$$

where $\mathcal{D}$ implies that the convergence is in distribution. Consequently, the distribution of the random variable on the left-hand-side can be approximated by the standard normal distribution for large enough n , that is,

$$\frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1) \quad \text{approximately asymptotically (i.e. for large } n\text{)}.$$

Similar to the notation adapted before, $N(0, 1)$ conventionally denotes the distribution of a standard normal random variable, sometimes it is used to represent a standard normal random variable itself.

Asymptotic pivotal quantity and confidence interval for sample mean

In the rare case that the variance σ^2 is known, if our interest is on inference of the population mean μ , consider

$$T_n = \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}}$$

Then T_n is an asymptotic pivotal quantity, in the sense that the asymptotic distribution of T_n depends on data and the unknown parameter μ only and the asymptotic distribution $N(0, 1)$ is parameter free. From

$$\mathbb{P}\left(z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = \mathbb{P}(z_{\alpha/2} \leq T_n \leq z_{1-\alpha/2}) \approx 1 - \alpha \quad \text{for large } n,$$

we can obtain an asymptotic confidence interval for μ ,

$$[L, U] = \left[\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}\right]$$

which has confidence level $\approx (1 - \alpha)100\%$.

Asymptotic confidence interval for sample mean (σ^2 unknown)

If our interest is on inference of the population mean μ while the population variance σ^2 is unknown as in most cases, the rearrangement from the probability statement of the asymptotic pivot

$$z_{\alpha/2} \leq T_n = \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} \leq z_{1-\alpha/2}$$

to the expression

$$L \leq \mu \leq U$$

will take more work and could be computationally demanding (e.g., the Wilson's method illustrated below).

An alternative approach is to replace the unknown σ^2 with further asymptotic approximation. This can be achieved by the Slutsky's Theorem.

Assume that $X_1, X_2, \dots$ are independent identically distributed random variables with mean μ and variance σ^2 , both μ and σ are unknown. Assume that there is a sequence of random variables U_n , with

$$U_n \rightarrow \sigma^2 \quad \text{in probability as } n \rightarrow \infty.$$

For example, by the Law of Large Numbers, the sample variance

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \rightarrow \sigma^2 \quad \text{in probability as } n \rightarrow \infty.$$

Recall that, when the limit is a constant, convergence in probability is the same as convergence in distribution.

Denote $\bar{X}_n$ as the sample mean of a sample of size n , and write

$$\frac{\bar{X}_n - \mu}{\sqrt{U_n/n}} = \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{Y_n}{W_n}$$

By the asymptotic normality and the convergence of U_n , the CLT and LLN yield

$$W_n \xrightarrow{\mathcal{P}} 1, \quad Y_n \xrightarrow{\mathcal{D}} Z, \quad \text{as } n \rightarrow \infty,$$

where $Z \sim N(0, 1)$, and the convergence is in probability and in distribution respectively.

Then, the Slutsky's Theorem claims

$$Y_n/W_n \xrightarrow{\mathcal{D}} Z \quad \text{as } n \rightarrow \infty.$$

In other words, the results of Slutsky's Theorem says that

$$T_n = \frac{\bar{X}_n - \mu}{\sqrt{U_n/n}} \sim N(0, 1) \quad \text{asymptotically.}$$

The asymptotic distribution of $N(0, 1)$ is free of parameters. If U_n are random functions of the data X_i 's only, then T_n is asymptotically pivotal. Therefore, when sample of size n is large, an asymptotic confidence interval

$$\mathbb{P}(L \leq \mu \leq U) = \mathbb{P}\left(\bar{X}_n - z_{1-\alpha/2}\sqrt{\frac{U_n}{n}} \leq \mu \leq \bar{X}_n + z_{1-\alpha/2}\sqrt{\frac{U_n}{n}}\right) \approx 1 - \alpha \quad (1)$$

can be derived from the normal approximation when the sample size n is large.

For example, if $\alpha = 0.05$, by the normal approximation, for large n , the coverage probability

$$\mathbb{P}(1.96 \leq T_n \leq 1.96) \approx \mathbb{P}(1.96 \leq Z \leq 1.96) = 0.95, \quad \text{where } Z \sim N(0, 1).$$

From which we can obtain an asymptotic 95% confidence interval for the population mean parameter μ ,

$$\left[\bar{X}_n - 1.96\sqrt{\frac{U_n}{n}}, \bar{X}_n + 1.96\sqrt{\frac{U_n}{n}} \right]$$

One of the most common choices is $U_n = S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, the sample variance of a sample of size n with mean $\bar{X}_n$, an unbiased estimator of σ^2 . Then the asymptotic 95% confidence interval for μ has the popular form

$$\left[\bar{X}_n - 1.96\sqrt{\frac{S_n^2}{n}}, \bar{X}_n + 1.96\sqrt{\frac{S_n^2}{n}} \right]$$

Note that, variance estimators other than S^2 (e.g., optimizations via MLE or MSE) are also common and useful.

Various methods based on asymptotic theory are developed to obtain confidence intervals, for example, the popular Wald confidence intervals, the score or Wilson confidence intervals, and the ones obtained by variance stabilization method.

4 Wald-type confidence intervals

Wald confidence interval is very popular, due to its simple structure and easy computation. In almost all statistical software, the default confidence interval in model output is by the Wald's method.

The name is after the Hungarian mathematician Abraham Wald.

We illustrate Wald's method for confidence intervals by examples.

Example (Wald confidence interval for the Poisson mean)

Data: $X_1, \dots, X_n$.

Model: X_i 's are *i.i.d.* $\sim \text{Poisson}(\lambda)$, that is, X_i 's are independent, with the probability mass function

$$P(X_i = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, \dots$$

Goal: Construct a Wald confidence interval for λ .

A few facts about the Poisson distribution:

- For X_i *i.i.d.* $\sim \text{Poisson}(\lambda)$, applying Taylor series result we can obtain $\mathbb{E}(X_i) = \lambda$, $\text{Var}(X_i) = \lambda$.
- The maximum likelihood estimator (MLE) of λ is $\hat{\lambda} = \bar{X}$, the sample mean.
- By the independence of the X_i , $\bar{X} \sim \text{Poisson}(\lambda)$.

By the Central Limit Theorem, when n is sufficiently large,

$$T_n = \frac{\bar{X} - \mathbb{E}(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \sim N(0, 1) \quad (\text{approximately})$$

The $N(0, 1)$ distribution is parameter free, therefore T_n is asymptotically approximately pivotal when n is large.

Wald's method is to produce a confidence interval in the form as stated in the expressions in (1), by replacing $\text{Var}(\bar{X})$ in the denominator with an estimator from data.

By the Law of Large Numbers,

$$\hat{\lambda} = \bar{X} \rightarrow \lambda \quad \text{in probability, as } n \rightarrow \infty$$

By the Slutsky's Theorem, when n is sufficiently large,

$$\frac{\hat{\lambda} - \lambda}{\sqrt{\hat{\lambda}/n}} \sim N(0, 1) \quad (\text{approximately})$$

which yields

$$\mathbb{P}\left(z_{\alpha/2} \leq \frac{\hat{\lambda} - \lambda}{\sqrt{\hat{\lambda}/n}} \leq z_{1-\alpha/2}\right) = \mathbb{P}\left(\hat{\lambda} - z_{1-\alpha/2}\sqrt{\frac{\hat{\lambda}}{n}} \leq \lambda \leq \hat{\lambda} + z_{1-\alpha/2}\sqrt{\frac{\hat{\lambda}}{n}}\right) \approx 1 - \alpha$$

Thus an asymptotic Wald-type confidence interval is

$$\left[\bar{X} - z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}}{n}}, \bar{X} + z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}}{n}} \right] = \left[\hat{\lambda} - z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + z_{1-\alpha/2} \sqrt{\frac{\hat{\lambda}}{n}} \right]$$

Example (Wald confidence interval for the Bernoulli - Binomial parameter)

Data: $X_1, \dots, X_n$.

Model: X_i 's are independent Bernoulli random variables, that is,

$$X_i = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1 - p \end{cases}$$

Goal: Estimate the parameter p and provide a confidence interval by the Wald's method.

Recall that previously, the exact pivotal method did not work. Now we turn to asymptotic methods.

When the sample size n is large (or moderate), the Central Limit Theorem asserts that

$$T_n = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\bar{X} - p}{\sqrt{p(1-p)/n}} \sim N(0, 1) \quad \text{asymptotically.}$$

The $N(0, 1)$ distribution does not depend on p , therefore again T_n is asymptotically approximately pivotal when n is large.

If we choose $\alpha = 0.05$, then $z_{1-\alpha/2} \approx 1.96$, then

$$\mathbb{P} \left(-1.96 \leq \frac{\sqrt{n}(\bar{X} - p)}{\sqrt{p(1-p)}} \leq 1.96 \right) \approx 0.95 \quad \text{for large } n.$$

From this probability expression, approximate 95% confidence intervals of parameter p can be derived (with some work).

Rewrite the above Normal asymptotic approximation as

$$\mathbb{P} \left(\bar{X} - 1.96 \sqrt{\frac{p(1-p)}{n}} \leq p \leq \bar{X} + 1.96 \sqrt{\frac{p(1-p)}{n}} \right) \approx 0.95$$

The problem is that the two bounds still depend on p . We aim to derive a confidence interval $[L, U]$ for parameter p , where the confidence interval limits L, U should depend on data and free of the unknown parameter p .

Wald's method corresponding to "plugging in" $\hat{p}$ in the place of p at the two ends of the inequalities in the above probability. We obtain the Wald's approximation

$$\mathbb{P} \left(\bar{X} - 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \bar{X} + 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right) \approx 0.95$$

The validity of the plug-in approximation is by using $\hat{p} \rightarrow p$ in probability as $n \rightarrow \infty$ (by the Law of Large Numbers) and then by applying the Slutsky's Theorem. Now the two interval endpoints are functions of data only. A 95% Wald-type asymptotic confidence interval for p can be derived as

$$\left[\bar{X} - 1.96 \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right] = \left[\hat{p} - 1.96 \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}, \hat{p} + 1.96 \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right]$$

Discussion (on Wald confidence intervals)

- Wald-type confidence intervals are of simple form and are the most popular.

- The Wald confidence interval for a population mean θ is symmetrically centered at $\hat{\theta} = \bar{X}$.

- In the Poisson and Bernoulli-Binomial examples,

What are the possible ranges of Wald confidence intervals?

What should be the ranges of the intervals?

- In the Bernoulli-Binomial example, if the observations are $X_i = 0$ for all $i = 1, \dots, n$:

What would be the value of the MLE $\hat{p}$, and what would be the Wald confidence interval for p ?

Analogously:

What would be the MLE and Wald confidence interval, if the observations are $X_i = 1$ for all $i = 1, \dots, n$?

- Wald confidence intervals can be problematic at borders of the parameter space.